



Process Quality Assurance (PQA)

PQA Overview

Required PA Information

Intent

Verifies and enables improvement of the quality of the processes performed and resulting work products.

Value

Increases the consistent use and improvement of the processes to maximize business benefit and customer satisfaction.

Additional Required PA Information

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Explanatory PA Information

Practice Summary

Level 1

PQA 1.1 Identify and address process and work product issues.

Level 2

PQA 2.1 Develop, keep updated, and follow a quality assurance approach and plan based on historical quality data.

PQA 2.2 Objectively evaluate selected performed processes and work products against the recorded process and applicable standards.

PQA 2.3 Communicate quality and non-compliance issues and ensure their resolution.

PQA 2.4 Record and use results of quality assurance activities.

Level 3

PQA 3.1 Identify and record opportunities for improvement during quality assurance activities.

Additional PA Explanatory Information

Objectivity in process quality assurance evaluations is critical to the success of the project. Evaluators should not evaluate their own work. Personnel independent from the project typically perform objective evaluations using defined criteria and a set of methods. These evaluations are a check of the performed processes and resulting work products against applicable process descriptions, standards, and procedures. Senior management takes an active role in process quality assurance by regularly reviewing results and taking action as needed.

A typical project applying quality assurance will:

- Demonstrate more consistent process implementation
- Have better insight into the project's results and issues
- Have better visibility into the project performance

Quality assurance verifies compliance with implementation to address issues such as:

- Incorrect and incomplete requirements
- Insufficient release planning
- Unresolved defects

Often, the verification and validation processes evaluate the same work products as quality assurance. Quality assurance focuses on determining that the verification and validation activities are done by following their recorded processes. Verification focuses on satisfaction of requirements. Validation confirms that the product works as intended in its target environment.

Promote an environment that encourages personnel to participate in identifying and reporting quality issues.

Related Practice Areas

Verification and Validation (VV)

Context Specific

Agile Development

Context Tag:	Agile Development
Context:	Integrate agile techniques and ceremonies with other processes.

Ensure that objective evaluations are integrated into the team's techniques or rhythms, e.g., as part of daily standups, story point estimation, code reviews, use of tools, continuous integration, and retrospectives.

An agile project has many opportunities to objectively evaluate ceremonies and work products, such as when:

- User stories are examined in the backlog grooming ceremony
- The Scrum Master coaches the team during scrum ceremonies
- Feedback on what was built is obtained in the Sprint review
- The retrospective ceremony reviews prior accomplishments and identifies opportunities for improvement for the next iteration
- Management or peers observe agile ceremonies being performed using techniques such as a gemba walk

Level 1

PQA 1.1

Required Practice Information

Practice Statement

Identify and address process and work product issues.

Value

Increases customer satisfaction through improved quality and performance.

Additional Required Information

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Explanatory Practice Information

Additional Explanatory Information

Identify issues that impede the project. Address these issues by either changing what is done or planning to make the changes the next time the process is performed. Identify work product defects and issues such as missing or incorrect information, formats that make it difficult to use the work product, or inconsistent use of terminology.

Example Activities

Example Activities	Further Explanation
Identify issues.	
Record issues.	
Resolve issues.	

Example Work Products

Example Work Products	Further Explanation
Recorded issues	
Addressed issues	Can be immediately addressed or changed in the future.

Level 2

PQA 2.1

Required Practice Information

Practice Statement

Develop, keep updated, and follow a quality assurance approach and plan based on historical quality data.

Value

Reduces cost and increases quality by focusing on recurring problem areas.

Additional Required Information

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Explanatory Practice Information

Additional Explanatory Information

Organizations may not have enough quality assurance resources to review everything in a project. Planning helps to optimize resource use and focus on the areas where the project is most likely to have process or work product issues. Using historical data helps to identify areas where applying resources will be most effective. Consider:

- Where issues have occurred in the past
- Quality trends
 - May be positive or negative. Instances where there have been few issues may not need quality resources applied to them, but areas where issues occur repeatedly may need increased quality activities.
- Sources of best practices
- Common or recurring problems across projects
- Recent changes in processes

Plan activities and determine the type and resources needed for objective evaluations:

- Ensure qualified personnel who perform quality assurance activities participate in developing plans, processes, standards, and procedures
- Identify and select processes and associated work products to be evaluated during the project
- Determine how and when quality assurance activities, findings, and results will be communicated and resolved within the organization

Example Activities

Example Activities	Further Explanation
Develop, record, and keep updated the quality approach and plan.	
Identify areas for evaluation.	
Review, update, and approve the approach with affected stakeholders.	

Example Work Products

Example Work Products	Further Explanation
Quality assurance approach and plan	Identify areas where the project lead would like objective feedback. May include: • Scope • Focus for objective evaluations including: ○ New processes ○ Processes with historical problems ○ Randomly selected processes • Work products to be objectively evaluated • Depth and coverage of the evaluation. For example: ○ In-depth evaluation ○ Observation ○ Quick review • Schedule ○ Date for the evaluation ○ Evaluation leader ○ Evaluation participants • Description of the quality assurance process, the reporting chain, and how objectivity will be ensured.

Related Practice Areas

Planning (PLAN)

PQA 2.2

Required Practice Information

Practice Statement

Objectively evaluate selected performed processes and work products against the recorded process and applicable standards.

Value

Delivers high-quality solutions by identifying and addressing issues throughout the process execution.

Additional Required Information

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Explanatory Practice Information

Additional Explanatory Information

Identifying and addressing quality issues should be done throughout the project and not just at the end. Catching and addressing issues early reduces the amount of rework. This also provides timely visibility into the quality of the processes and the resulting products. Objectivity is critical to the success of quality assurance evaluations. Critical processes or work products may require independence from the work performed as a way of achieving objectivity.

Objectivity can be achieved by using:

- Independent quality assurance organizations or groups
- Independent reviewers who:
 - Are not involved in development or maintenance of the solution
 - Work in a separate management reporting chain
- Criteria such as standards, guidelines, etc.
- Checklists based on process descriptions, standards, and procedures

Examples of objective evaluation methods include:

- Formal audits
- Peer reviews with objective reviewers, which can be performed at various levels of formality
- In-depth review of work at the place it is performed, e.g., desk audits
- Distributed review and comment regarding work products
- Built-in or automated process checks to identify incorrectly performed processes, e.g., Poka-yoke

Example Activities

Example Activities	Further Explanation
Develop and keep updated clearly stated criteria for evaluations.	
Develop and keep updated checklists based on process descriptions, standards, and procedures.	
Use the defined criteria and checklists to evaluate if selected performed processes follow process descriptions, standards, and procedures.	
Identify and record each noncompliance found during the evaluation.	
Leverage recorded best practices in other parts of the organization.	Submit improvement proposals for the best practices.

Example Work Products

Example Work Products	Further Explanation
Criteria	
Checklists	
Evaluation reports	
Noncompliance reports	
Improvement proposals	

Context Specific

Data

Context Tag:	CMMI-DATA
Context:	Use processes to incorporate data management best practices as an integral part of the solution.

Ensure objective evaluation of work products, e.g., criteria and requirements, to verify and enforce objectivity, and to incorporate all aspects of managing data, e.g., consistent use of the data glossary terms, data architecture.

Safety

Context Tag:	CMMI-SAF
Context:	Use processes to incorporate safety considerations as an integral part of the solution, work, project, and organization.

Ensure objective evaluation of work products, e.g., criteria and checklists, incorporate all aspects of safety activities, e.g., safety objectives, approach to address workplace environment safety, approach to address functional safety, organizational safety function, safety evaluations, and organizational safety controls.

Security

Context Tag:	CMMI-SEC
Context:	Use processes to incorporate security considerations as an integral part of the solution, work, project, and organization.

Ensure objective evaluation of work products, e.g., criteria and checklists, incorporate all aspects of security activities, e.g., security objectives, approach to address security in the workplace environment, approach to address physical security needs, mission security needs, and cybersecurity.

PQA 2.3**Required Practice Information****Practice Statement**

Communicate quality and non-compliance issues and ensure their resolution.

Value

Ensures quality processes, avoids the cost of rework, and improves customer satisfaction.

Additional Required Information

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Explanatory Practice Information**Additional Explanatory Information**

Noncompliance issues are problems identified when team members do not follow applicable standards, recorded processes, or procedures. The status of noncompliance issues tracked over time provides an indication of quality trends. Address and resolve noncompliance issues in the project. Take actions to resolve non-compliance issues at the level closest to the occurrence of the non-compliance whenever possible. Escalate noncompliance issues through the management chain for resolution when needed.

Example Activities

Example Activities	Further Explanation
Communicate and resolve each noncompliance issue.	Consider reporting a compliance percentage rather than the number of noncompliance issues for a project. Report quality assurance activities, results, and issues to senior management on a regular basis so they can take timely and appropriate action. Ways to resolve noncompliance issues include: • Fixing the noncompliance • Changing the applicable recorded processes, standards, or procedures • Obtaining a waiver to cover the noncompliance issue and accept the associated risk Track noncompliance issues to resolution.
Escalate noncompliance issues when they cannot be resolved.	Escalation may continue through multiple layers of management levels until the issue is resolved.
Analyze noncompliance issues to identify quality trends.	These trends can then be used to focus future quality activities.
Ensure affected stakeholders are aware of evaluation results and quality trends.	

Example Work Products

Example Work Products	Further Explanation
Quality trend analysis reports	
Noncompliance resolutions	

PQA 2.4

Required Practice Information

Practice Statement

Record and use results of quality assurance activities.

Value

Optimizes future quality assurance activities and reduces the cost of future work.

Additional Required Information

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Explanatory Practice Information

Additional Explanatory Information

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Example Activities

Example Activities	Further Explanation
Record and keep updated information about quality assurance activities.	Record information about quality assurance activities with enough detail to clarify status and results. Noncompliance issues recorded as part of the peer review report are tracked and escalated outside the project when necessary.

Example Work Products

Example Work Products	Further Explanation
Evaluation records	
Quality assurance reports	
Status reports of non-compliance issues and corrective actions	
Reports of quality trends	

Level 3

PQA 3.1

Required Practice Information

Practice Statement

Identify and record opportunities for improvement during quality assurance activities.

Value

Improves the organization's capability to meet its goals and objectives.

Additional Required Information

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Explanatory Practice Information

Additional Explanatory Information

Quality assurance includes:

- Evaluating the process as performed
- Identifying ways that the process can be improved
- Submitting improvement proposals

Quality assurance personnel should work closely with process management to ensure an efficient and effective process is deployed, followed, and kept updated. Through this relationship, the process is continuously maintained and improved.

Example Activities

Example Activities	Further Explanation
Record potential improvements observed during quality assurance activities.	Includes: • Suggested process changes • Observations about effectiveness • Related activities which may or may not be a part of the current organizational process
Submit improvement proposals.	

Example Work Products

Example Work Products	Further Explanation
Improvement proposals	

